BOUNDED COMPLEXES OF PERFECT MODULES ARE PERFECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Two-out-of-three for perfect objects: in a distinguished triangle, if two terms are perfect then so is the third. invoked at proof lines 18982-18984. Closes the induction step by passing from the truncated complex and the perfect module to the full complex.

Source reference. Original lemma 066R: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18949–18949. Cited at proof lines 18982-18984 to close the induction step; the two-out-of-three property is load-bearing inside the counted span.

Lemma 0.2. *Let R be a ring. Let $K^\bullet$ be a bounded complex of perfect R-modules. Then $K^\bullet$ is a perfect complex.*

Proof. Follows by induction on the length of the finite complex: use Lemma 066R and the stupid truncations. □

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